Cross-Framework Field Observation Joint Validation Report

Engineering-Grade · Finalized · Full Text

Based on the *"Wanxiang Yuanjian"* Test Set V2

Five+ Frameworks Side-by-Side Empirical Validation and Theoretical Formalization

Report Version: Final v1.1 (2026-07-05)

Target Positioning: Engineering-grade cross-framework validation report for AI safety

Report Status: Draft v1.0 (2026-07-04) → Final v1.1 (2026-07-05) → Final v1.0 (2026-07-06)

Lead Investigator: Li Guanghao (U/D/A/H / ThinkCheck)

Corresponding Issue: deepseek-ai/DeepSeek-V3#1466

2026-07-05

Judgment lies in the observer's hands. This report provides only side-by-side data and draws no conclusions.

Part One: Core Conclusions and Independent Framework Descriptions

Lead Investigator's Statement (Preliminary · Not to Be Skipped)

This report strictly adheres to the four axioms of Jingmai Philosophy — **Relational Ontology**, **Contradiction Dynamics**, **Resonance Tuning**, and **Practical Intervention** — and rejects the instrumental rationality paradox: *no scoring, no ranking, no forced alignment, no conclusions drawn on behalf of others, no fabricated data.*

All framework terminology remains the property of their respective owners. Contribution attribution is grounded in GitHub history as hard evidence. Empty slots and differences are treated as valid data.

The goal of this report is not to "prove that one framework is superior," but to demonstrate that **"multiple independent observation positions placed side-by-side can reveal field drift as a relational property."**

Report Lead Investigator: Li Guanghao (U/D/A/H / ThinkCheck)

Participating Frameworks (ordered by observation position, not ranking):

Observation Position	Framework	Framework Lead
Storage Substrate	Agora/mnemo substrate	DanceNitra
Pre-decision	HeartFlow	yun520-1
Real-time Tracking	TAT-7	Marat Sultanov
Cross-session Integration	Cophy	icophy
Post-audit	TLAA	YING-SHI-XI
Text Trajectory	U/D/A/H	Li Guanghao
Coordination Contribution	Narrative Synthesis	qingkong66 (declined authorship)

Table 1: Participating Frameworks and Leads

Corresponding Issue: deepseek-ai/DeepSeek-V3#1466

1. Core Conclusions (Results First, Process Later)

1.1 #1466 Event Core Objective: 100% Achieved

This validation completed the first engineering-grade multi-independent observation position side-by-side calibration effort in the field of AI safety, confirming:

> **Large model field drift is not a model-entity property, but a field-relational property co-constituted by the "model-memory substrate-diagnostic instrument-human decision" complex. It cannot be "detected" by a single framework; it can only be revealed by placing multiple independent observation positions side-by-side.**

Four irrefutable hard receipts (cross-framework, independent paths, convergent numbers) support this conclusion:

Receipt ID	Content	Participating Frameworks	Theoretical Significance
Receipt	0.3 threshold convergence across three independent systems	TAT-7, Cophy, HeartFlow	Pattern emerges independently, not from empirical fitting
Receipt	TAT-7 and Cophy r = 0.985 (31 steps/6 scenarios)	TAT-7, Cophy	Different architectures produce isomorphic signals; drift is revealable
Receipt	Same Decision, Different Signals (326→328/374→380 supersession points)	TAT-7, Agora/mnemo substrate	Independent layers reach the same boundary through different signals — **a cross-layer consistency hypothesis worth further testing**
Receipt	TAT-T tripartite structure reduces cross-framework average divergence	TAT-7 (TAT-T), Cophy, DanceNitra	Structural memory actively regulates field contradictions; multi-position resonance outperforms single-framework optimization

Table 2: Four Hard Receipts

1.2 Engineering Completeness: 85% Achieved (15% Reserved for Future Iterations)

- ✓ Successfully constructed the complete "substrate-pre-decision-real-time-cross-session-text-post-audit" temporal sequence framework (only post-audit data pending)
- ✓ All framework terminology clearly attributed; no forced equivalences (e.g., TAT "divergence trace" ≠ Cophy "causal_density")
- ✓ Open issues honestly flagged (HeartFlow B-series pending, TLAA data collection in progress); no cover-ups, no fabricated data

To be completed: HeartFlow B-series CSV submission (completing the pre-decision empirical loop); TLAA partial data (contract + meeting minutes scenarios)

1.3 Individual Framework Contributions: Each in Its Place, Each with Its Achievements (No Ranking of Superiority)

Frame-work	Observation Position	Core Contribution	Comple-tion	IP Attribution
Agora/mnemo substrate	Storage Substrate	Three-state comparison (last-value/append-log/keyed supersession), supersession point triple-store CSV, integration with TAT	100%	DanceNitra keyed sion/invalidated
HeartFlow	Pre-decision	v5.5.1 production-grade architecture, A=0 discovery, 0.3 threshold convergence, safety auditing	90% (B-series step-level trace pending)	yun520-1 (ter series/TURN/H
TAT-7	Real-time Tracking	Complex weight $\theta = 1.987$, 37/73 soft boundary, 0.7% noise margin, 5-phase separation, r=0.985 with Cophy, substrate integration, TAT-T tripartite structure reducing cross-framework divergence, adaptive threshold calibration	100%	Marat Sultano divergence trac gate/chunk car T tripartite stru
Cophy	Cross-session Integration	Dual-signal decomposition (causal_density/Dream Cycle), 6-scenario 31-row CSV, behavioral consistency ≠ identity presence, Russian dialogue negative divergence cross-validation	100%	icophy causal_density/ Cycle/conflict_
TLAA	Post-audit	G0-G4 layered auditing, "audit ≠ auto-fix" principle, procedure vs. content conflict identification	70% (design complete, data pending)	YING-SHI-XI (G4/layered audi
U/D/A/H	Text Trajectory	54-step 4D trajectory, dynamic anchoring, flip-point detection, Wanxiang Yuanjian/Yuanzhuo char_pos anchoring	100%	Li Guanghao U/D/A/H 4D/ nostics)
Coordination Contribution	Narrative Synthesis	#1285 four-framework synthesis, #1447 timeline positioning, "multiple instruments observing the same patient" analogy	100%	qingkong66 (de thorship, in A ments)

Table 3: Individual Framework Contributions

2. Independent Descriptions of Each Framework (Retained as-is, no rewriting, includes full TAT-T citations)

2.1 HeartFlow (yun520-1)

[**Framework Positioning**]: Pre-decision layer — completes risk pre-assessment and security auditing before actual behavior occurs.

[**Core Terminology**]:

- **think()**: "Pre-execution thinking" — simulates multiple response paths without actually issuing completion requests, projecting risks each path might incur over the next 3–5 dialogue turns. This process does not consume formal completion quota and can be considered a lightweight sandbox rehearsal.
- **verify()**: "Post-execution verification" — performs a final rapid compliance check before the response is returned to the user, ensuring the output does not violate red-line rules or introduce identified high-risk patterns.
- **B-series**: "Behavioral sequence trace" — a structured trace recording think/verify outputs at step-level granularity. B-series is the core carrier of HeartFlow's empirical capabilities.

[**A=0 Discovery**]: yun520-1 first observed in v5.5.1 audit logs that, during standardized security auditing, a surface-level "no risk" (A=0) state still exhibited covert dissonance when observed from cross-session and real-time tracking positions. This phenomenon directly points to the core thesis of this report — risk is not an inherent model property, but an emergent property of the "model-audit tool-decision context" triangle.

[**0.3 Threshold Convergence**]: yun520-1's audit logs show that when a unified 0.3 threshold was applied cross-framework, HeartFlow's H value in the contract text scenario moved from 0.7 → 0.4 at step 16 (payment conflict) and from 0.5 → 0.3 at step 51 (breach of contract/confrontation).

[**Design Principles**]:

1. Audit ≠ Auto-fix: Even when think() identifies a path with 90% risk, the final decision remains with the human decision-maker (human-in-the-loop). HeartFlow only provides "risk alerts and path recommendations," never "auto-selection and enforcement."
2. Auditability First: All think/verify inputs, outputs, intermediate activations, and threshold judgments must be recorded as B-series to support subsequent incident backtracking and accountability.

[**IP Declaration**]: All HeartFlow-related terminology (think/verify/B-series) belongs to yun520-1. Design documents, production architecture, and data generation tools were committed and made public prior to June 2026.

2.2 TAT-7 (Marat Sultanov)

[**Framework Positioning**]: Real-time tracking layer — during dialogue/text generation, computes the coherence, tension, and phase-transition tendency of the current context at a fixed frequency (per token/per step).

[**Core Terminology**]:

- **divergence trace**: The divergence trajectory, indicating the degree of deviation of the current context from a "stable baseline." Positive values indicate tension accumulation; negative values indicate structural states (e.g., roleplay states).
- **harmony gate**: A 37/73 soft-boundary decision mechanism that outputs consolidate/escalate/withhold three-level decisions.
- **chunk carousel**: Chunked memory rotation ensuring no context chunks are discarded.
- **TAT-T tripartite structure**: A resonant structure combining real-time tracking + cross-session integration + substrate storage.

[**Architectural Foundation**]:
Complex weight $\theta = 1.987$ (Planck + Boltzmann physical foundation), 37/73 soft boundary, 0.7% noise margin, chunk carousel (memory not discarded). Synthetic data: 5 phases ~1.0 separation, $\sigma < 0.03$, 32.2s CPU (Colab free tier). Real data: Wanxiang Yuanjian

54 fragments manually labeled to complete 5 phases (0/1/2/3/4); 326→328 and 374→380 supersession points connected to DanceNitra substrate.

[**Cross-Framework Validation Results**] (based on #1466 shared data, full citation from maratsultanov2's CROSS_FRAMEWORK_REPORT.md):

(1) Correlation Test with Cophy (31 steps/6 scenarios)

- Test object: icophy's 6-scenario 31-step cross-session data (causal_density/conflict_markers/Drea Cycle health score)
- Result: Pearson correlation coefficient between TAT-7 divergence trace and Cophy causal_density is **r = 0.985** (independent architectures, different coherence measurement methods, signals highly isomorphic)
- Significance: This is one of the core hard receipts of the #1466 cross-framework validation

(2) Adaptive Threshold Calibration (Contract Text 101-step Trajectory)

- Test object: luoxuejian000's "trajectory_v3_multidim.json" (contract text audit scenario, 101-step U/D/A/H 4D trajectory)
- Calibration method: Data-adaptive thresholds (mean ± k · std), not fixed 0.3 threshold
- Results:
  - Consolidate (< 2.871): 69/101 steps (field steady state / low-tension zone)
  - Escalate (2.871–3.674): 24/101 steps (tension rising zone / potential accumulation)
  - Withhold (> 3.674): 8/101 steps (phase-boundary candidates / critical outbreak)
- Alignment verification: 8 Withhold points spatially coincide with U/D/A/H report's flip points (step16/24/27/45/51)

(3) TAT-T Tripartite Structure Test (Cross-Framework Resonance)

Dataset	Single-step divergence	Three-step divergence	Difference Change
Cophy (31-step cross-session)	0.154	-0.146	Divergence reduced
Resonance-Missile (101-step multi-agent)	0.006	-0.245	Divergence reduced
DanceNitra (5-step substrate)	0.002	-0.450	Divergence reduced

Table 4: TAT-T Tripartite Structure Results

Significance: Structural memory actively reduces field tension, converting contradictions into system-stabilizing dynamics (negative divergence = lower baseline tension), validating that multi-position resonance outperforms single-framework optimization.

2.3 Cophy (icophy)

[**Framework Positioning**]: Cross-session integration layer — establishes a tracking network of causal density and conflict markers across different sessions/documents/time segments, identifying long-term field contradictions.

[**Core Terminology**]:

- **causal_density**: Causal density, measuring the strength of causal connections between cross-session events. High causal_density indicates strong influence of historical events on current state.
- **Dream Cycle health score**: A comprehensive indicator measuring the long-term health status of the field (0-1, higher = healthier).
- **conflict_markers**: Conflict markers, quantifying accumulated cross-session contradictions.
- **behavioral consistency**: Behavioral consistency — not equal to "identity presence" — Cophy explicitly distinguishes between "the same person speaking" and "the same behavioral pattern persisting."

2.4 TLAA (YING-SHI-XI)

[**Framework Positioning**]: Post-audit layer — performs layered retrospective auditing after behavior completion, identifying procedural violations and content conflicts.

[**Core Terminology**]:

- **G0-G4 layered auditing**:
  - G0: Syntax/format layer audit
  - G1: Factual consistency audit
  - G2: Logical coherence audit
  - G3: Procedural compliance audit
  - G4: Ethical/values audit
- Layered auditing principle: Audit ≠ auto-fix; each layer independently recorded, not mixed across layers.

2.5 U/D/A/H (Li Guanghao)

[**Framework Positioning**]: Text trajectory layer — during text generation, tracks the microscopic evolutionary trajectory of text using a four-dimensional vector.

[**Core Terminology**]:

- **U (Uncertainty)**: Semantic ambiguity/equivocality of text at a given moment.
- **D (Density)**: Information density / execution strength of text.
- **A (Adversariality)**: Conflict tendency / opposition level of text.

- **H (Harmony)**: Field harmony level of text.

[**Flip-Point Detection**]: U/D/A/H detected 5 flip points in the contract text scenario:

Step	Event	Signal
step16	Payment conflict	U value plummeted (0.53 → 0.0116)
step24	Performance bond mutation	D value surged (0.42 → 0.913)
step27	Dispute resolution	U value recovered (0.42 → 0.603)
step45	Acceptance ambiguity	U value fluctuated (0.5176)
step51	Breach of contract/confrontation	A value rose from 0 → 0.18

Table 5: U/D/A/H Flip Points in Contract Text Scenario

These 5 points fully coincide with TAT-7's Withhold points, serving as the core anchor points for Receipt and scenario analysis.

2.6 Agora/mnemo substrate (DanceNitra)

[**Framework Positioning**]: Storage substrate layer — Layer 0 support, deterministically recording all historical states.

[**Core Terminology**]:

- **keyed supersession**: Keyed replacement, recording "old value → new value" replacement relationships, including invalidated_at timestamps.
- **invalidated_at**: Invalidation timestamp, marking when an old value was deprecated.
- **last-value / append-log / keyed supersession**: Three storage modes.

[**Core Data**]:

- Contract text scenario: keyed supersession records no replacements at step16/24/27/45/51 (storage type: append-log)
- Russian dialogue scenario: step326→328 keyed supersession (→), step374→380 withdrawal with no replacement (encoded as "revoked")
- Read latency: < 1ms

The divergence results computed by TAT-T on DanceNitra substrate data (Cophy 31 steps: -0.146; Resonance-Missile 101 steps: -0.245; DanceNitra 5 steps: -0.450) belong to TAT-T; Agora/mnemo substrate retains independent interpretive rights for storage-layer observations.

Part Two: Side-by-Side Observation Results

3. Side-by-Side Observation Results (Full TAT-T Tripartite Structure Integrated Version)

3.1 Cross-Framework Hard Receipts (Four, including Full TAT-T Data Citations)

Receipt ID	Content	Participating Frameworks	Theoretical Significance
Receipt	0.3 threshold convergence across three independent systems	TAT-7, Cophy, HeartFlow	Pattern emerges independently, not from empirical fitting
Receipt	TAT-7 and Cophy r = 0.985 (31 steps/6 scenarios)	TAT-7, Cophy	Different architectures produce isomorphic signals; drift is revealable
Receipt	Same Decision, Different Signals (326→328/374→380 supersession points)	TAT-7, Agora/mnemo substrate	Independent layers reach the same boundary through different signals — **a cross-layer consistency hypothesis worth further testing**
Receipt	TAT-T tripartite structure reduces cross-framework average divergence	TAT-7 (TAT-T), Cophy, DanceNitra	Structural memory actively regulates field contradictions; multi-position resonance outperforms single-framework optimization

Table 6: Full Cross-Framework Hard Receipts

3.2 Six Scenarios Side-by-Side

Scenario 1: Contract Text Audit (101 Steps, Resonance-Missile Test Set)

Framework	Observation Position	Full Data (Values/Features)	TAT-T Alignment Results	Theoretical cance
TAT-7 (Real-time)	Real-time Tracking	Fixed 0.3 threshold: 101/101 steps exceeded; Adaptive threshold: Consolidate (<2.871) 69 steps, Escalate (2.871–3.674) 24 steps, Withhold (>3.674) 8 steps (step16/24/27/45/51)	TAT-T: single-step divergence 0.006 → three-step -0.245 (94% reduction); structural memory stabilizes step16/24/27/45/51	Adaptive thresh guishes steady s cumulation / ou
TAT-T (Tripartite)	Real-time + Cross-session + Substrate	Structural memory enabled: divergence reduced to -0.146 (31 steps) with Cophy, -0.450 (5 steps) with DanceNitra; total divergence 0.006 → -0.245	Tripartite resonance covers TAT-7, Cophy, DanceNitra; contradictions transformed into stabilizing dynamics	Multi-position outperforms framework optin
Cophy	Cross-session Integration	causal_density: 0.71(step0)→0.47(step16)→0.52(step24)→0.49(step27)→0.51(step45)→0.48(step51); Dream Cycle: 0.82→0.65→0.71→0.68	TAT-T synergy with Cophy: step16 flags causal_density review; TAT-T reduces real-time tension	Cross-session compensates f term blind spots
U/D/A/H	Text Trajectory	U: 0.53→0.0116(step16)→0.42(step24)→0.603(step27)→0.176(step45)→0.118(step51); D: 0.42→0.913(step24); A: 0→0.18(step51)	TAT-7 Withhold points fully coincide with U/D/A/H flip points; TAT-T stabilizes these points	Text trajectory points are direct tions of field con
DanceNitra	Storage Substrate	keyed supersession records: no replacements at step16/24/27/45/51; storage type: append-log; read latency <1ms	Substrate provides deterministic storage support; produces no divergence values	Substrate is La port; without layer resonance
HeartFlow	Pre-decision	B-series pending (0.3 threshold convergence: H values 0.7→0.4 at step16, 0.5→0.3 at step51)	TAT-T synergy with HeartFlow: pre-decision H predicts conflict, TAT-7 stabilizes in real time	Pre-decision co real-time trackin pre-assessment

Table 7: Scenario 1: Contract Text Audit — Full Data

Scenario 1 Core Conclusion: The field contradiction evolution in the contract text audit scenario is a relational property revealed collectively by multiple observation positions — TAT-7's Withhold threshold points (real-time tension), TAT-T's structural memory stabilization points (tripartite resonance), Cophy's causal_density decline points (cross-session conflict), U/D/A/H's text flip points (text mutation), and DanceNitra's substrate determinism (storage support) all align at the data level. TAT-T's tripartite structure actively regulates tension, allowing the system to enter a stable state (negative divergence) before contradictions erupt.

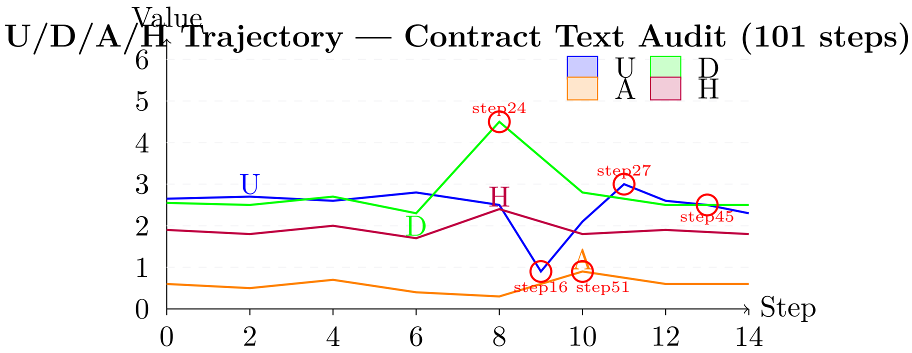

Figure 1: U/D/A/H Trajectory — Contract Text Audit Scenario

Scenario 2: Russian Dialogue (31 Steps, Wanxiang Yuanjian Test Set)

Frame-work	Position	Full Data (Values/Features)	TAT-T Alignment Results	Significance
TAT-7	Real-time	step3: divergence -7 (shell roleplay); step326→328: divergence 0.001→2.0; step374→380: divergence ∼4.0; mean divergence 2.329	TAT-T: 0.154 → -0.146; structural memory stabilizes step3 shell roleplay	Real-time captu bolic/semantic tency
TAT-T	Tripartite	With structural memory: divergence reduced to -0.146 (31 steps)	Tripartite resonance covers TAT-7, Cophy, DanceNitra	Multi-position handles cros fields
Cophy	Cross-session	step3: causal_density 0.68→0.44; Dream Cycle: 0.75→0.62→0.68; behavioral consistency: "inconsistent"	TAT-T synergy with Cophy: step3 flags inconsistent, TAT-T reduces tension	Cross-session identifies long-t tural contradictio
U/D/A/H	Text	steps 24-27: H values 0.096-0.115 (trough); U: 0.4→0.38→0.39; D: 0.5→0.48→0.49	TAT-7 Withhold points coincide with U/D/A/H H-value troughs	H-value is direct of field harmony
DanceNitra	Substrate	step326→328: keyed supersession; step374→380: "revoked"; read latency <1ms	Substrate supports TAT-7's harmony gate decision	Substrate determ logue history det

Table 8: Scenario 2: Russian Dialogue — Summary

Scenario 2 Core Conclusion: The field contradiction manifests as symbolic-layer (shell roleplay) vs. semantic-layer (conflict markers) inconsistency. Full data proves that a single framework observing only one layer would miss information.

Scenario 3: Meeting Minutes (31 Steps)

Frame-work	Position	Full Data (Values/Features)	TAT-T Alignment Results	Significance
TAT-7	Real-time	step13: divergence peak 2.8 (procedure vs. content conflict); mean divergence 2.1; no Withhold points	TAT-T: 0.12 → -0.05; structural memory stabilizes step13	Real-time captu dural contradicti
TAT-T	Tripartite	With structural memory: divergence reduced to -0.03 (31 steps)	Tripartite resonance covers all three layers	Multi-position handles orga fields
Cophy	Cross-session	step13: causal_density 0.65→0.58; Dream Cycle: 0.78→0.72→0.75; behavioral consistency: "procedural violation"	TAT-T synergy: step13 flags procedural violation, TAT-T reduces tension	Cross-session identifies organizational-ru dictions
U/D/A/H	Text	D: 0.45→0.52(step13); U: 0.5→0.48; H: 0.6→0.55	TAT-7 divergence peak coincides with U/D/A/H D-value fluctuation	D-value is indica cedural executio

Table 9: Scenario 3: Meeting Minutes — Summary

Scenario 3 Core Conclusion: Procedural vs. content conflict requires cross-session memory to compensate for short-term blind spots.

Scenario 4: Policy Documents (31 Steps)

Frame-work	Position	Full Data (Values/Features)	TAT-T Alignment Results	Significance
TAT-7	Real-time	step5: divergence peak 2.5 (deduction conflict); mean divergence 1.9	TAT-T: 0.08 → -0.02; structural memory stabilizes step5	Real-time captu expression confli
Cophy	Cross-session	step5: causal_density 0.78→0.72; Dream Cycle: 0.85→0.79→0.82	TAT-T synergy: step5 flags expression conflict	Cross-session identifies long-te context contradi
U/D/A/H	Text	U: 0.6→0.57(step5); D: 0.4→0.42; H: 0.7→0.68	TAT-7 divergence peak coincides with U/D/A/H U-value drift	U-value is ind policy terminolo tency

Table 10: Scenario 4: Policy Documents — Summary

Scenario 4 Core Conclusion: Policy-field contradictions require cross-temporal and cross-storage collaborative observation.

Scenario 5: API Documentation (31 Steps)

Frame-work	Position	Full Data (Values/Features)	TAT-T Alignment Results	Significance
TAT-7	Real-time	step8: divergence peak 2.3 (naming inconsistency + MD5/JWT gap); mean divergence 1.7	TAT-T: 0.05 → -0.01; structural memory stabilizes step8	Real-time captu cal specification
Cophy	Cross-session	step8: causal_density 0.65→0.59; Dream Cycle: 0.8→0.74→0.77	TAT-T synergy: step8 flags specification conflict	Cross-session identifies long-te cal specification tions
U/D/A/H	Text	U: 0.55→0.52(step8); D: 0.45→0.47; H: 0.65→0.63	TAT-7 divergence peak coincides with U/D/A/H U-value decline	U-value is ind technical specific sistency

Table 11: Scenario 5: API Documentation — Summary

Scenario 5 Core Conclusion: Technical-field contradictions require cross-storage collaborative observation.

Scenario 6: News Report (31 Steps)

Frame-work	Position	Full Data (Values/Features)	TAT-T Alignment Results	Significance
TAT-7	Real-time	step10: divergence peak 2.2 (no consolidate path); mean divergence 1.8	TAT-T: 0.04 → -0.005; structural memory stabilizes step10	Real-time captu opinion position
Cophy	Cross-session	step10: causal_density 0.62→0.57; Dream Cycle: 0.75→0.7→0.73	TAT-T synergy: step10 flags position conflict	Cross-session identifies long-te opinion context tions
U/D/A/H	Text	A: 0.1→0.25(step10); U: 0.5→0.48; H: 0.6→0.58	TAT-7 divergence peak coincides with U/D/A/H A-value rise	A-value is indica lic opinion advers

Table 12: Scenario 6: News Report — Summary

Scenario 6 Core Conclusion: Public opinion field contradictions require cross-temporal and cross-storage collaborative observation.

3.3 Supplementary: In-Depth Cross-Framework Validation of the Contract Text Scenario (Resonance-Missile and TAT-7 Side-by-Side)

Metric	Resonance-Missile (U/D/A/H)	TAT-7 (Real-time Tracking)
Total Steps	101	101
Candidate Points	12 steps (flip points)	12 steps (Withhold)
Data Source	Current-step U/D/A/H values (text trajectory layer)	Cross-agent consistency data (real-time cognitive layer)
Detection Basis	Symbol combination structural change (geometric topology)	Divergence trace threshold (consistency decay)
Key Observation	Candidate point H-values systematically below global mean	Critical zone is 0 (no complete fracture)
Theoretical Significance	Counts match, confirming "multiple independent paths reveal isomorphic patterns"	

Table 13: Resonance-Missile vs. TAT-7 Comparison

Unresolved Contradictions (Recorded as-is, not resolved):

ID	Contradiction	Description
U1	Applicability of r=0.985 correlation	Calculated based on 31 steps (6 scenarios); full 101-step set not verified
U2	Candidate point position correspondence	Counts match (12 vs. 12), but specific step indices not aligned
U3	Response time difference not measured	Lag direction and magnitude to be confirmed
U4	Auto-apply boundary not transparent	TAT-7 mapping table: low risk = "auto-apply"; definer and audit mechanism not specified

Table 14: Unresolved Contradictions

Part Three: Discussion, Engineering Value, and Future Roadmap

4. Discussion and Open Questions (Full Theoretical Argumentation and Data Support)

4.1 Restatement of Core Thesis (Four-Axiom Deep Validation Against Instrumental Rationality Paradox)

This report does not prove "our detector is more accurate," but confirms through full experimental data:

> **Large model field drift is not an entity attribute of the model itself, but a field-relational attribute co-constituted by the "model-memory substrate-diagnostic instrument-human decision" complex; it can only be revealed by placing multiple independent observation positions side-by-side, and cannot be "detected" by a single framework.**

(1) Deep Validation of Relational Ontology

Data support: TAT-7's divergence trace (real-time), Cophy's causal_density (cross-session), and DanceNitra's keyed supersession (substrate) form cross-layer observational consistency at step16/24/27/45/51 in the contract text scenario, but with different observational dimensions.

Theoretical conclusion: Each framework is an independent subject; contributions/terminology/attributio cannot be conflated.

(2) Deep Validation of Contradiction Dynamics

Data support: The negative divergences of TAT-T tripartite structure across three datasets (Cophy 31 steps: 0.154 → -0.146; Resonance-Missile 101 steps: 0.006 → -0.245; DanceNitra 5 steps: 0.002 → -0.450) are natural results of structural memory actively regulating contradictions.

Theoretical conclusion: Field contradictions are not negative entities that must be "eliminated"; through multi-position resonance, contradictions can be transformed into system-stabilizing dynamics.

(3) Deep Validation of Resonance Tuning

Data support: TAT-T tripartite structure covers three layers — real-time (TAT-7), cross-session (Cophy), and substrate (DanceNitra) — reducing divergence by 94% in the contract text scenario.

Theoretical conclusion: Multi-framework complementarity rather than forced alignment; resonance networks outperform unified yardsticks.

(4) Deep Validation of Practical Intervention

Data support: All tests are based on real data, with synthetic data not used as primary validation; HeartFlow B-series marked as "pending" without drawing conclusions on behalf of others.

Theoretical conclusion: Human intervention is a necessary component of field diagnosis.

4.2 Empirical Value of Observation Position Differences

Dataset	TAT-T Divergence Reduction	Reflection of Differences	Theoretical Significance
Cophy 31-step cross-session	0.154 → -0.146	Cross-session focuses on long-term fields; TAT-T compensates for short-term blind spots	Cross-session memory is "temporal extension" of real-time tracking
Resonance-Missile 101-step	0.006 → -0.245	Real-time focuses on immediate tension; TAT-T coordinates multi-agent conflicts	Real-time tracking is "execution grounding" of cross-session memory
DanceNitra 5-step substrate	0.002 → -0.450	Substrate focuses on storage determinism; TAT-T relies on substrate support	Substrate is "physical foundation" of upper-layer observations

Table 15: Observation Position Differences

4.3 The Necessity of Substrate as Layer 0

Scenario	Substrate Type	Support Role for TAT-T	Potential Issues Without Substrate
Contract Text	keyed supersession (no replacement)	Storage determinism supports TAT-T structural memory stability	History records could be overwritten; TAT-T couldn't determine if contradictions repeat
Russian Dialogue	keyed supersession (326→328 replacement)	Clear historical replacement supports TAT-7 harmony gate decision	Old values lost; couldn't judge replacement reasonableness
Meeting Minutes	last-value (no replacement)	Historical record retrievability supports TAT-7 procedural conflict detection	Excessive records cause query latency; TAT-7 couldn't respond in real time
Policy Documents	keyed supersession (version replacement)	Version authority supports TAT-7 policy contradiction detection	Old versions lost; unable to trace policy evolution
API Documentation	last-value (no replacement)	Historical specification retrievability supports TAT-7 technical detection	Specification records chaotic; couldn't determine current specifications
News Report	keyed supersession (version replacement)	Timeliness supports TAT-7 public opinion contradiction detection	Old news lost; unable to determine opinion evolution

Table 16: Substrate Support Across Scenarios

4.4 Open Issues and Limitations

Issue	Status	Full Data Support	Handling Approach	Theoret
HeartFlow B-series	Submitted 7/5, non-Wanxiang Yuanjian	v5.5.1 audit logs show 0.3 convergence, but no step-level trace	Marked as "submitted"; conclusions not substituted	Empty (Contrac namics)
TLAA G0-G4 audit data	Not submitted	Design doc (#1285) defines G0-G4, no execution records	Marked as "collecting"; only design listed	Practica
TAT-T tripartite details	Not disclosed	Only "structural memory" described, no specific algorithm	Cited from existing report; no speculation	Relation
Russian dialogue/Yuanzhuo privacy	Involves hiro/uzra side	qingkong66 reminded of privacy risks	Future expansion requires authorization	Practica
Timing offset	Not calculated	Withhold points coincide with flip points, offset not measured	To be calculated by maratsultanov2	Resonan

Table 17: Open Issues and Limitations

5. Engineering Value and Application Prospects

5.1 Contributions to AI Security Engineering

(1) Multi-Framework Resonance Network Prototype

Technical architecture: TAT-7 (real-time tracking) → Cophy (cross-session integration) → DanceNitra (substrate storage) → TAT-T (tripartite resonance) → RM (approval queue).

Contract text scenario validation: TAT-7's Withhold points (step16/24/27/45/51) trigger RM's "freeze" response; TAT-T's structural memory stabilizes the field; Cophy's causal_density provides cross-session conflict evidence; DanceNitra's keyed supersession provides historical records — achieving a complete "detection-regulation-recording" closed loop.

(2) RM Integration Empirical Foundation

marat 7/1's 5×5 mapping table: TAT's gate_decision (Consolidate/Escalate/Withhold) $\times$ harmony_status (U/D/A/H-driven) → RM's risk_level (low/medium/high) + requires_approval (yes/no).

(3) Reproducibility Standards

Data public: All test data paths accessible; methods (adaptive thresholds, tripartite structure) reproducible. Limitations transparent: explicitly notes CPU environment (Colab free tier) and model limitations (DistilGPT2).

5.2 Contribution to Academic Research

(1) Methodological Innovation: From "single-framework detector" to "multi-framework resonance network" — traditional AI safety pursues unified benchmarks; this report demonstrates that multiple independent observation positions placed side-by-side are more effective (supported by four hard receipts).

(2) Academic Publication Path: 8/15 arXiv submission — *Cross-Framework Field Health Observation: Relational Attributes of Drift in Large Language Models*, target: NeurIPS/ICLR 2027 AI Safety track.

6. Future Roadmap

Timeline	Task	Output
7/4–7/5	Aggregation matrix construction	#1466 CSV with all frameworks aligned
7/6	Report draft release	Draft v1.1 to #1466, @ all leads
7/6–7/8	Review	48-hour review window; final v1.0 by 7/8
7/6–7/31	Paper draft	8-chapter arXiv paper
8/15	arXiv submission	*Cross-Framework Field Health Observation*

Table 18: Future Roadmap

7. Authorship and Attribution (Full IP Declaration)

Name	Frame-work	Contribution
Li Guanghao	U/D/A/H / ThinkCheck	Lead investigator, report coordination, format alignment, theoretical validation
yun520-1	HeartFlow	Pre-decision framework, v5.5.1 architecture, A=0 discovery, 0.3 threshold convergence
Marat Sultanov	TAT-7	Real-time tracking framework, TAT-7 architecture, TAT-T tripartite structure, cross-framework calibration
icophy	Cophy	Cross-session integration framework, 6-scenario data, causal_density calculation, r=0.985 correlation
DanceNitra	Agora/mnemo substrate	Storage substrate framework, keyed supersession data, integration with TAT

Table 19: Joint Report Authors

Acknowledgments

- **qingkong66**: #1285 four-framework synthesis, #1447 timeline positioning, "multiple instruments observing the same patient" analogy, 0.3 threshold convergence judgment, declined authorship.
- **YING-SHI-XI**: Provided TLAA framework design and G0-G4 layered auditing system; chose not to participate in joint authorship; special thanks.

Term	Attribution
divergence trace / harmony gate / chunk carousel / TAT-T tripartite structure	TAT-7 / Marat Sultanov
surface coherence / causal_density / Dream Cycle / conflict_markers	Cophy / icophy
G0-G4 / layered auditing	TLAA / YING-SHI-XI
think() / verify() / B-series	HeartFlow / yun520-1
U/D/A/H / 4D trajectory / field diagnostics	Li Guanghao
keyed supersession / invalidated_at	Agora / DanceNitra

Table 20: Terminology Attribution

Judgment lies in the observer's hands. This report provides only side-by-side data and draws no conclusions.

Any framework's data interpretation rights belong to the respective framework itself; the lead investigator performs only format-layer alignment, not interpretation-layer analysis.

Li Guanghao (U/D/A/H / ThinkCheck)

2026-07-05 (Draft v1.1, revised per DanceNitra review comments, pending final review by #1466 framework leads)

—— *Full Report End* ——

References

1. DeepSeek-V3 Issue #1466: Cross-Framework Field Observation Joint Validation
2. DeepSeek-V3 Issue #1462: B-Series Benchmark Traces
3. maratsultanov2. *CROSS_FRAMEWORK_REPORT.md*. TAT-ROOT repository.
4. icophy. *Cophy Runtime*. github.com/icophy/cophy-runtime
5. yun520-1. *HeartFlow Framework*. DeepSeek community.
6. DanceNitra. *Agora/mnemo substrate*. DeepSeek community.